

Dhirubhai Ambani University

MA614 – Preparatory Mathematics

MSc Agriculture Analytics, Semester 1

Course File

Instructor: Dr. Bibin Baby John

Credits: 2 — Hours: 30 (approx)

Academic Year: 2025–2026

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Course Information

Course Title:	MA614 – Preparatory Mathematics
Programme:	M.Sc. Agriculture Analytics (Sem 1)
Credits:	2
Total Contact Hours:	24-28
Instructor:	Dr. Bibin Baby John
Institute:	Dhirubhai Ambani University
Academic Year:	2025-2026

Course Description

This course provides foundational mathematical concepts required for M.Sc. Agriculture Analytics students who have studied mathematics up to secondary (10th) level. The emphasis is on geometric intuition, concrete examples from agriculture and simple computational skills to prepare students for further coursework in analytics and statistics.

Course Objectives

1. Reinforce essential mathematical ideas required for quantitative courses in analytics.
2. Provide geometric and intuitive understanding of vectors, functions, calculus and probability.
3. Equip students with basic skills to interpret and manipulate simple mathematical models relevant to agricultural data.

Course Outcomes (COs)

CO	Course Outcome	Bloom's Level
CO1	Understand basic ideas of vectors, matrices, and solving linear equations.	Understand (L2)
CO2	Describe and apply sets, relations and functions in everyday data contexts.	Apply (L3)
CO3	Interpret rates of change and areas using calculus ideas with geometric intuition.	Understand (L2)
CO4	Explain simple probability concepts and compute basic probabilities.	Apply (L3)

1. Syllabus and Module Plan

Overview

Total hours: 24-28. Credits: 2.

Modules (tentative-30 hours)

Module	Topics	Hours	Emphasis
1. Linear Algebra (Basics)	Vectors as arrows; representation of points; linear equations; 2x2 matrices and operations; solving small linear systems by elimination; visual intuition for span and direction.	8	Visual + computational
2. Sets, Relations, Functions	Set operations, Venn diagrams, Cartesian product, relations and properties, mapping and functions, domain and range, plotting simple functions.	5	Visual examples
3. Calculus (Elementary)	Concept of limit intuitively; derivative as slope/rate; simple derivative rules and interpretation; increasing/decreasing; integration as area under curve (Riemann idea with rectangles); basic applications.	8	Geometric intuition + computational
4. Probability (Introductory)	Random experiments; sample space; events; axioms (informal); simple/combinatorial probability; conditional probability and independence (basic intuition); expectation (simple cases).	7	Applications in agriculture (rainfall, germination)

2. Teaching–Learning Methods

- Blackboard/Openboard demonstrations with step-by-step geometric sketches.
- PPT slides as class notes, and for summary and graphs
- In-class short exercises and discussions using agricultural examples (crop yields, rainfall patterns, simple surveys).

3. Assessment Plan

Component	Weight	Description
Continuous Assessment (Quizzes/Assignments)	20%	Frequent short quizzes after modules and class participation. One class assignment.
Mid-sem/Internal Test	30%	Covers Modules 1–2 (first half of syllabus).
End-Semester Exam	50%	Cumulative exam with short problems and interpretation questions.

4. Lecture-wise Plan (tentative)

Part I – Before Mid-semester

Lec	Topic	Activity / Learning Outcome
1	Vectors and Introduction to linear algebra, linear combination, span	Geometric sketches; practice adding vectors.
2	Linear equations and solving 2×2 systems (elimination)	Step-by-step solving examples.
3	Matrix as a transformation; apply to linear systems	Hands-on solving.
4	Determinants, cofactors, minor, and Inverse	Geometric intuition and practice.
5	Ranks, independence	Geometric intuition and practice.
6	Eigen values and eigen vectors	Geometric intuition and practice.
7	Orthogonality and projection	Geometric intuition and practice.
8	Set operations, Venn diagrams, and cartesian product	Group activity: represent simple surveys.
9	Relations and functions; domain and range, types	Sketching maps and function tables.
10	Functions – introduction, types	Plot examples by hand.
11	Functions – algebra and another classification	Plot examples by hand.
12	Mid-term exam	Modules 1 and 2.

Part II – After Mid-semester

Lec	Topic	Activity / Learning Outcome
13	An intuitive introduction to calculus	Visual approach.
14	Derivative as slope and basic rules	Rate examples (growth, speed).
15	Applications: maxima/minima basics	Simple optimization examples.
16	Integration as area; rectangles approximation	Hands-on area estimate.
17	Application - Area between curves, Average value of a function, Physical applications	Agricultural examples -practice
18	Probability basics: experiments and outcomes	Coin/ball experiments.
19	Conditional probability and independence	Agricultural examples (disease given moisture).
20	Expectation and simple distributions	Compute expected values.
21	Practice set	Preferred agriculture related examples
22	Quiz and discussion	Discussion.
23	Revision: mixed problems	Group solving.
24	Test / Final review	Short test or cumulative Q&A.

5. References and Learning Resources

- Gilbert Strang, Linear Algebra and Its Applications
- David C. Lay, Linear Algebra and Its Applications
- Sheldon Axler, Linear Algebra Done Right
- Jim Hefferon, Linear Algebra (open-access) —
- Elementary Calculus by Michael Corral
- OpenStax Precalculus / Calculus (free online resources).
- NCERT Class XI, XII Mathematics (for basic revision notes and exercises).
- Schaum's Outlines (Linear Algebra / Calculus) – short practice problems.
- Sheldon Ross, An Introduction to Probability.
- Mendenhall, Intro to Probability & Statistics.